AI101 MINI-PROJECT

AQI Hazardous Prediction

Classifying hazardous air-quality days in Indian cities from 12 pollutant readings with Logistic Regression.

GitHub github.com/2026aib1072-dev/Air-quality-ai101-project

TEAM AND ROLES

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Problem and data

WHAT

Flag hazardous air-quality days in Indian cities from 12 pollutant readings, and show which pollutants drive the call.

WHY

Long exposure to PM2.5, PM10, CO and SO2 harms lungs and hearts. An automatic flag supports early warnings.

24,850	12	74 : 26
daily records used	pollutant features	safe : hazardous days

Source: Kaggle, Air Quality Data in India (city_day.csv). 29,531 raw rows × 16 columns; 24,850 rows carry an AQI label.

12 INPUT FEATURES

PM2.5 · PM10 · NO · NO2 · NOx · NH3 · CO · SO2 · O3 · Benzene · Toluene · Xylene

Indian AQI scale: our label splits it at 200. Target column: Is_Hazardous.

Non-hazardous (0): AQI ≤ 200

Good	Satisfactory	Moderate
0–50	51–100	101–200

Hazardous (1): AQI > 200

Poor	Very Poor	Severe
201–300	301–400	401–500

Approach

Data

1. **Load and clean**
   29,531 rows → 24,850 with an AQI value
2. **Create the label**
   Is_Hazardous = 1 when AQI > 200
3. **Stratified 80/20 split**
   19,880 train · 4,970 test, class ratio kept
4. **Impute and scale**
   Median fill + scaling, fit on train only

Model

Logistic Regression

scikit-learn, max_iter = 1000

Returns the probability that a day is hazardous.

Output

- Accuracy, F1, log loss
- Confusion matrix
- Odds ratio per pollutant
- Borderline days: model unsure (40–60% probability)

Why Logistic Regression? Binary target, a probability for every day, and coefficients that convert into odds ratios.

Built with Python · Pandas · NumPy · scikit-learn · Matplotlib · Seaborn · Jupyter · Git and GitHub

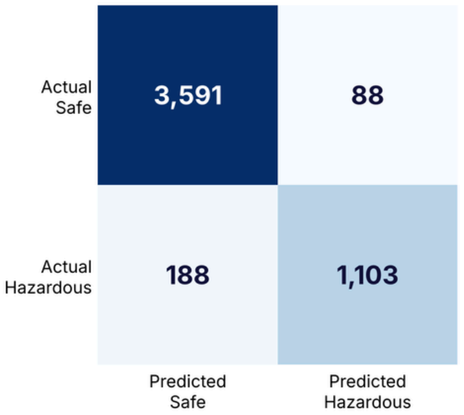
Results

94.45%
test accuracy (train: 94.36%)

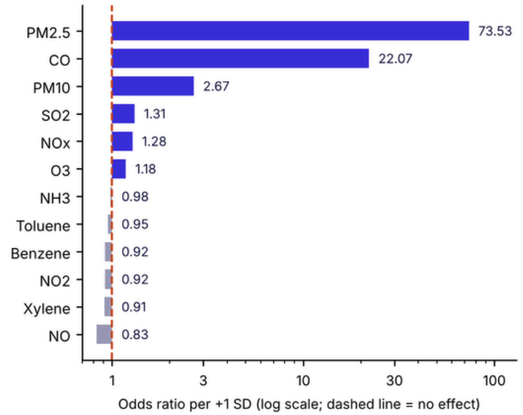
0.889
F1-score on hazardous days

0.85
recall on hazardous days (precision 0.93)

174
borderline days: model unsure (40–60% probability)



Confusion matrix, 4,970 test days



Odds ratio per +1 SD, log scale

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2. Tipping Point Analysis

Found 174 borderline days on the edge of hazardous classification.

Average Raw Pollutant Levels on Borderline Days:
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	Average_Concentration
PM2.5	83.3042
PM10	154.6888
NO	19.8882
NO2	31.5133
NOx	39.9849
NH3	22.0043
CO	2.0932
SO2	15.4584
O3	41.4251
Benzene	2.8311
Toluene	7.3879
Xylene	1.6639

Notebook output: borderline-day analysis

How to read it
PM2.5 and CO drive hazardous days: odds ×73.5 and ×22.1. · No over- or underfitting: train 94.36%, test 94.45%. · 188 of 1,291 hazardous days were missed.

Learnings

What worked

- Train 94.36% vs test 94.45% accuracy: no over- or underfitting
- Imputer and scaler fit on training data only: no leakage
- Odds ratios explain the model: PM2.5 ×73.5, CO ×22.1

What did not

- Hazardous recall is 0.85: 188 of 1,291 hazardous days missed
- AQI is built from the same pollutants, so high accuracy is partly expected
- Heavy imputation for sparse columns such as Xylene; one random split, no cross-validation

Who did what

Vineet
Data loading, cleaning, split · README

Rishabh
EDA, initial model · requirements.txt, run steps

Divya
Evaluation and plots · video

Sandeep
Odds ratios, borderline analysis · slides

AI-use disclosure

ChatGPT: explaining ML concepts and code troubleshooting.
Claude: polishing the README, helping draft requirements.txt .
All code, results and interpretation are ours. Every number comes from our notebook runs on city_day.csv.